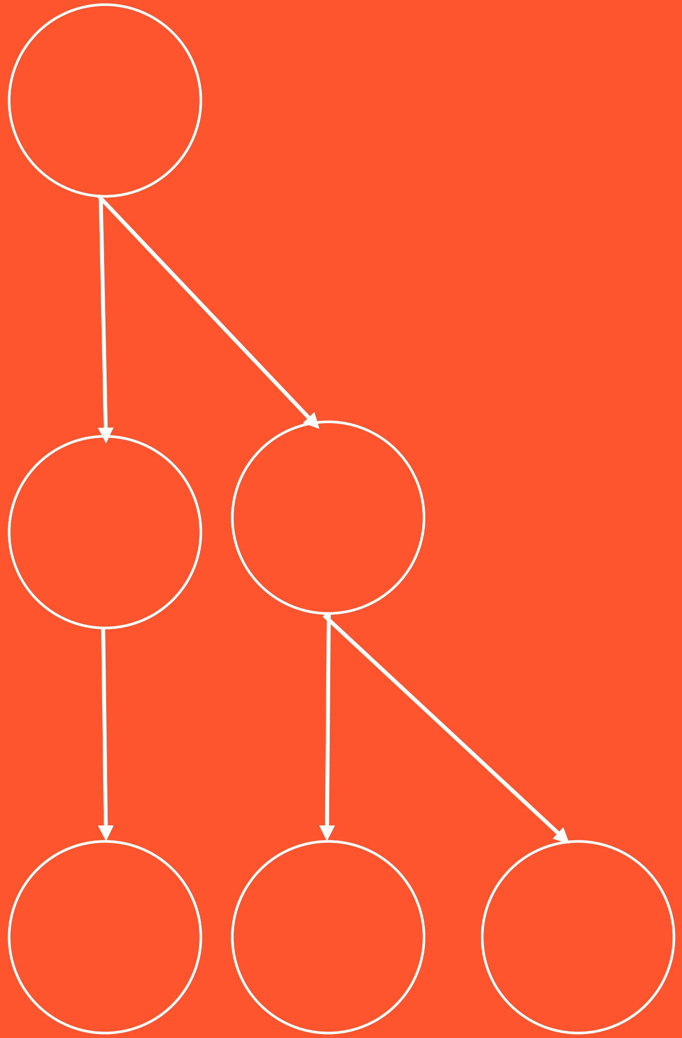


BCOG 100

Pattern Recognition

Lecture 1: The Problem of Pattern Recognition

UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN

Reading Check 7-1

Password: pattern

Question: what is connectionism?

The Pattern Recognition Problem

Computation
What does the system do?

Algorithm
How does the system do it?

Implementation
How is the system realized?

Recap & Overview

Brains that learn

Today:

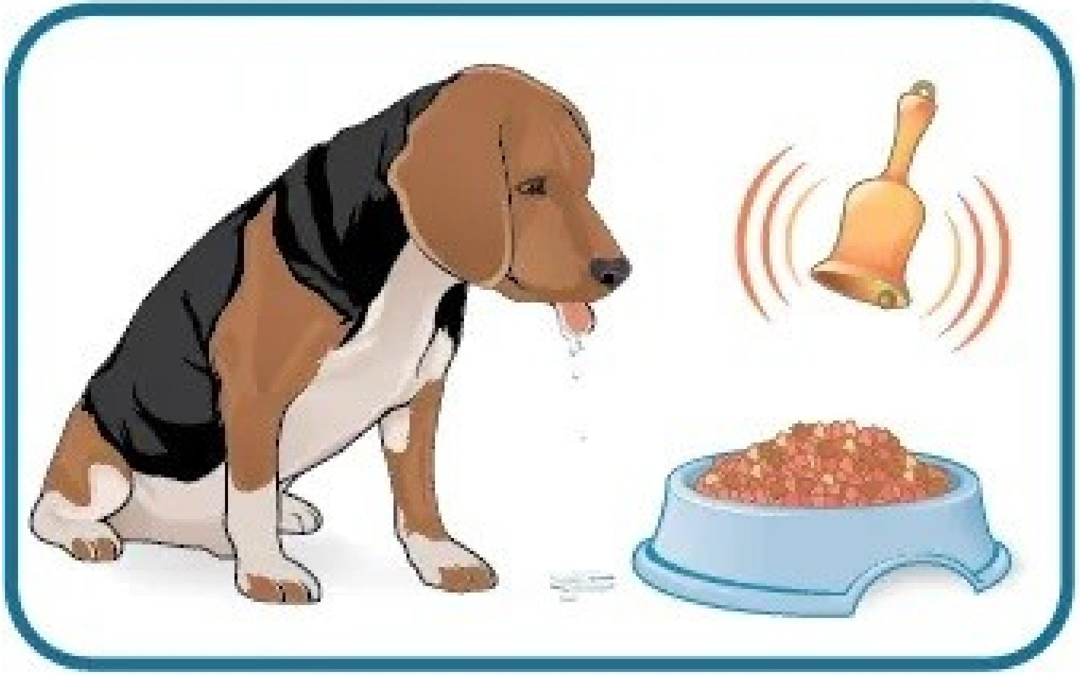
- From learning to perception and action
- From associations to structured recognition
- Pattern recognition as the bridge between sensing and acting

The Pattern Recognition Problem

From Reflex to Pattern Recognition

- Early bilaterians relied on reflex arcs
- Learning added flexibility but stayed local
- Recognition integrates cues into structure
- Chaotic Cambrian environments
- Overlapping and unreliable signals
- Survival demanded detecting meaningful regularities

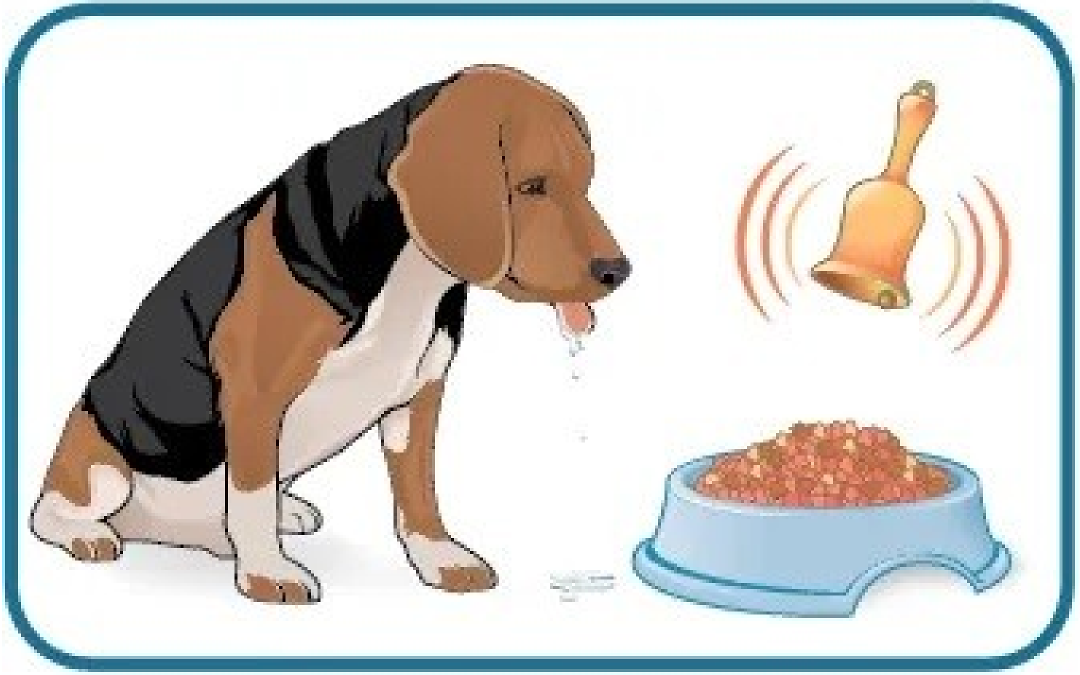
The Pattern Recognition Problem



Learning Meets Perception

- Hebbian learning linked co-activation
- Reinforcement learning assigned value
- Both required recognizable input patterns
- But the world is noisy and ambiguous
- Single cues mislead or fluctuate
- Associative links fail under variation

The Pattern Recognition Problem



The Core Pattern-Recognition Problem

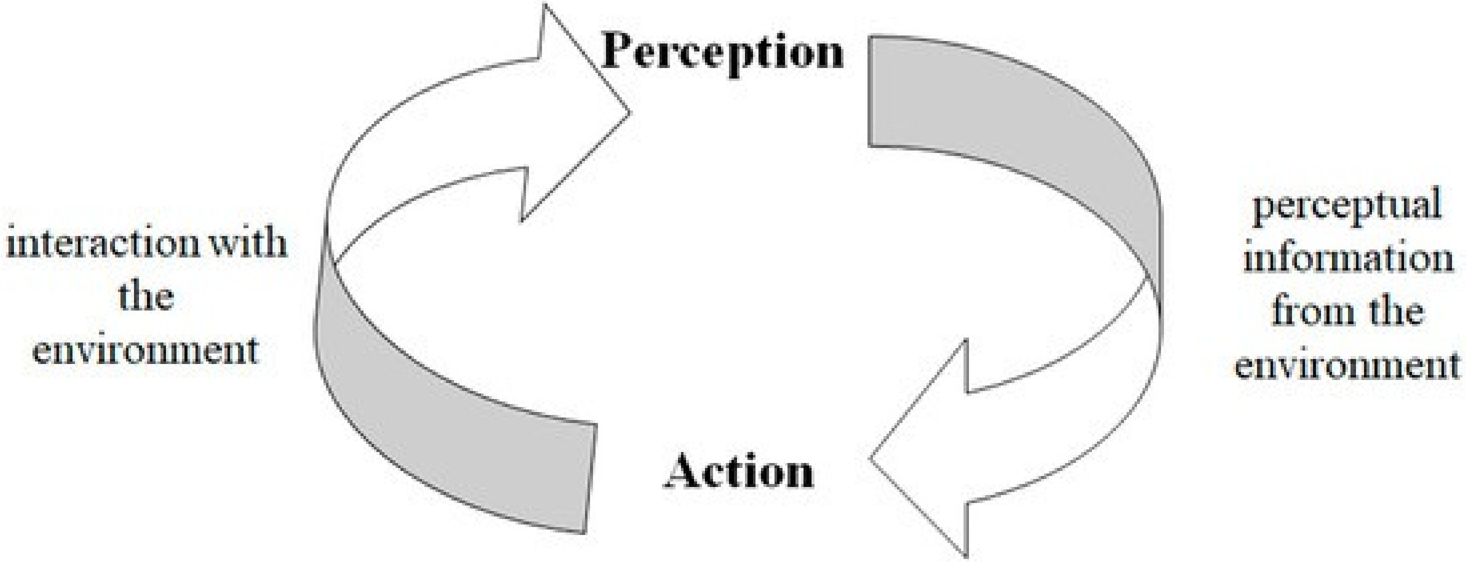
- Distinguish meaningful from irrelevant input
- Extract stability from variation
- Map patterns to appropriate actions

The Pattern Recognition Problem

What Pattern Recognition Is

- Detect regularities in sensory input
- Map them to internal states or categories
- Guide adaptive, context-sensitive behavior

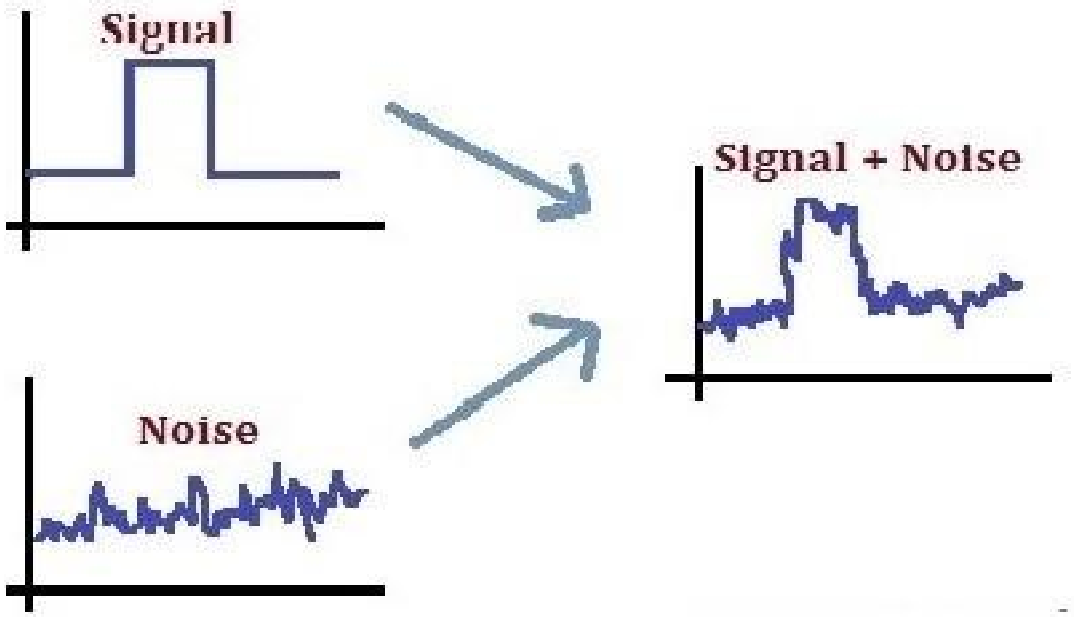
The Pattern Recognition Problem



Perception and Action

- Recognition informs movement
- Movement changes sensory input
- Continuous feedback supports adaptation

The Pattern Recognition Problem



Signal and Noise

- Natural signals are unreliable
- Neurons fire stochastically
- Decisions must be made under uncertainty

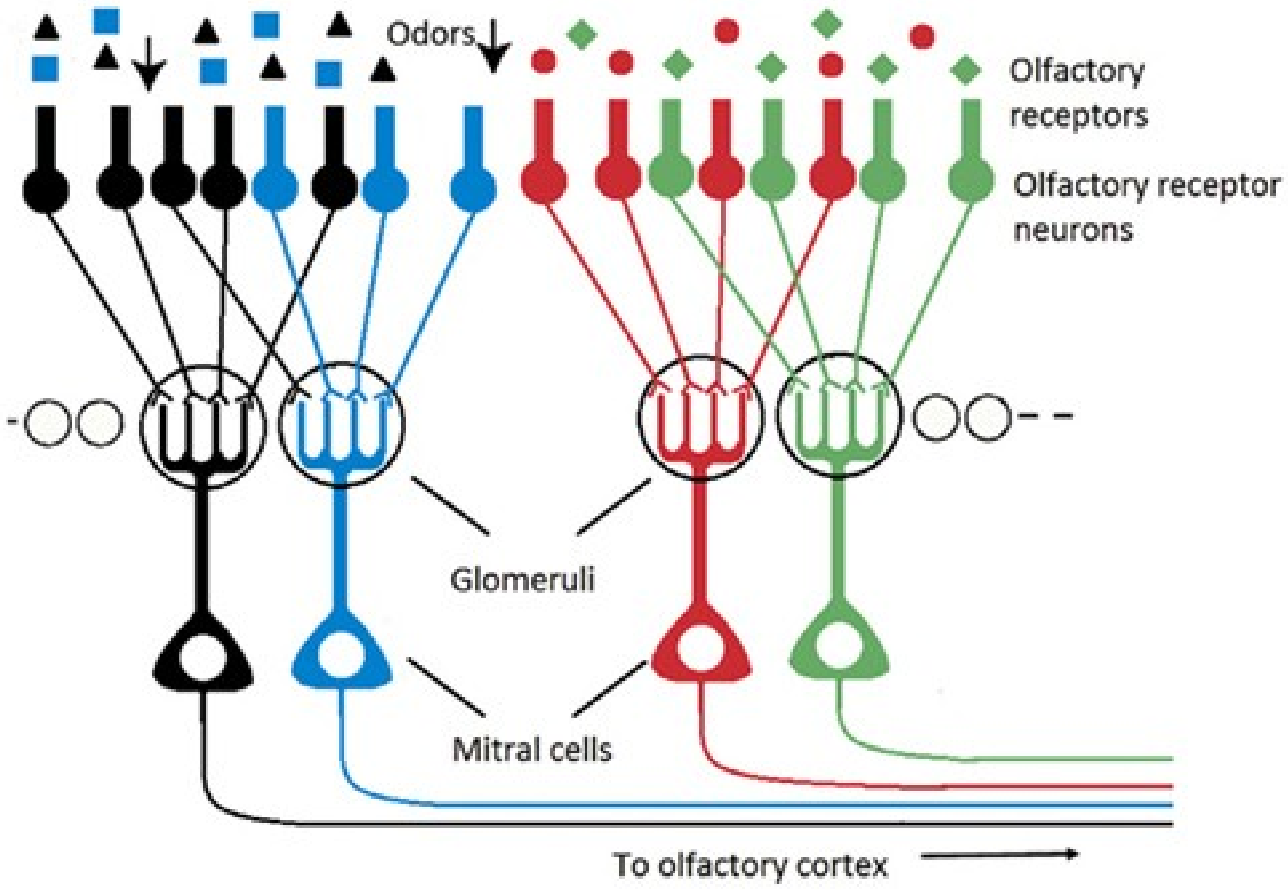
The Pattern Recognition Problem

		Stimulus There for Real	
		Yes	Non
Guess That Stimulus Was There	Yes	Hit	False Alarm
	No	Miss	Correct Rejection

Signal Detection Theory

- Hit, miss, false alarm, correct rejection
- Balance sensitivity and specificity
- Ecology sets the optimal trade-off

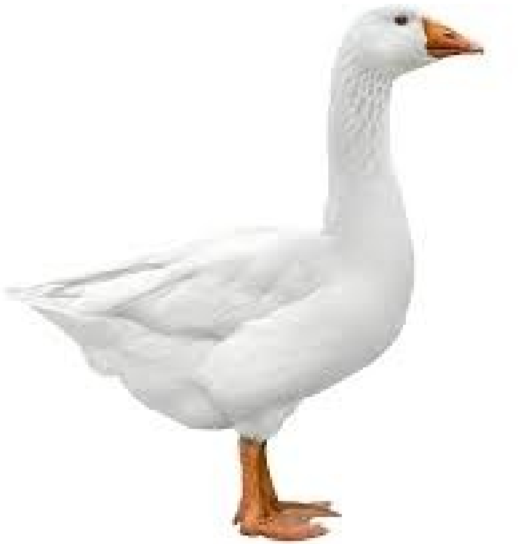
The Pattern Recognition Problem



Biological Noise Reduction

- Population coding averages variability
- Converging neurons boost reliability
- Statistical pooling increases accuracy

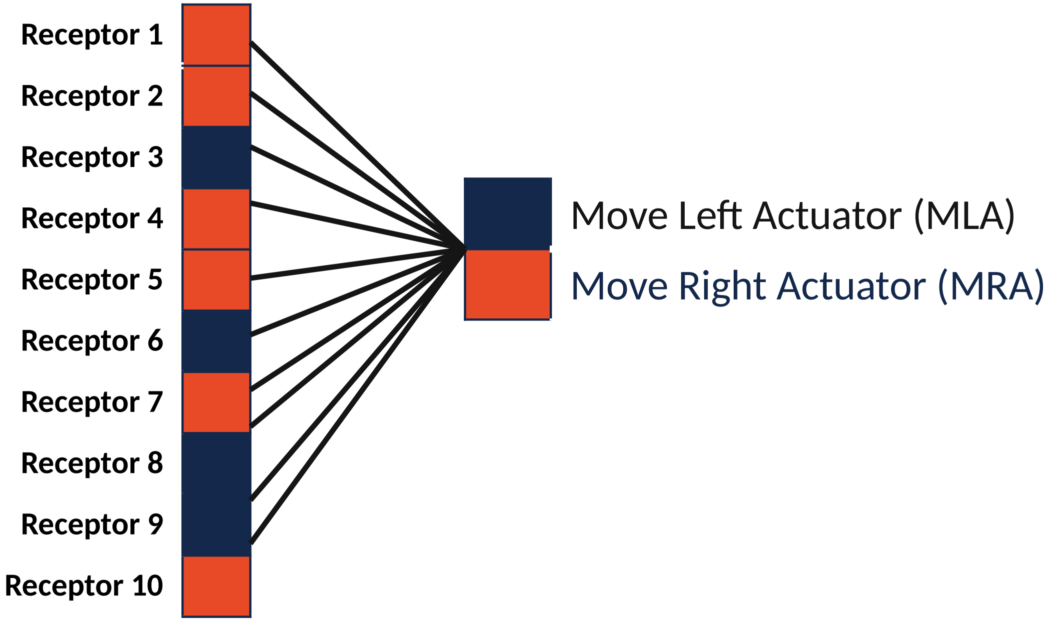
The Pattern Recognition Problem



Discrimination and Generalization

- Discrimination separates categories
- Generalization groups variants
- Both are essential for survival

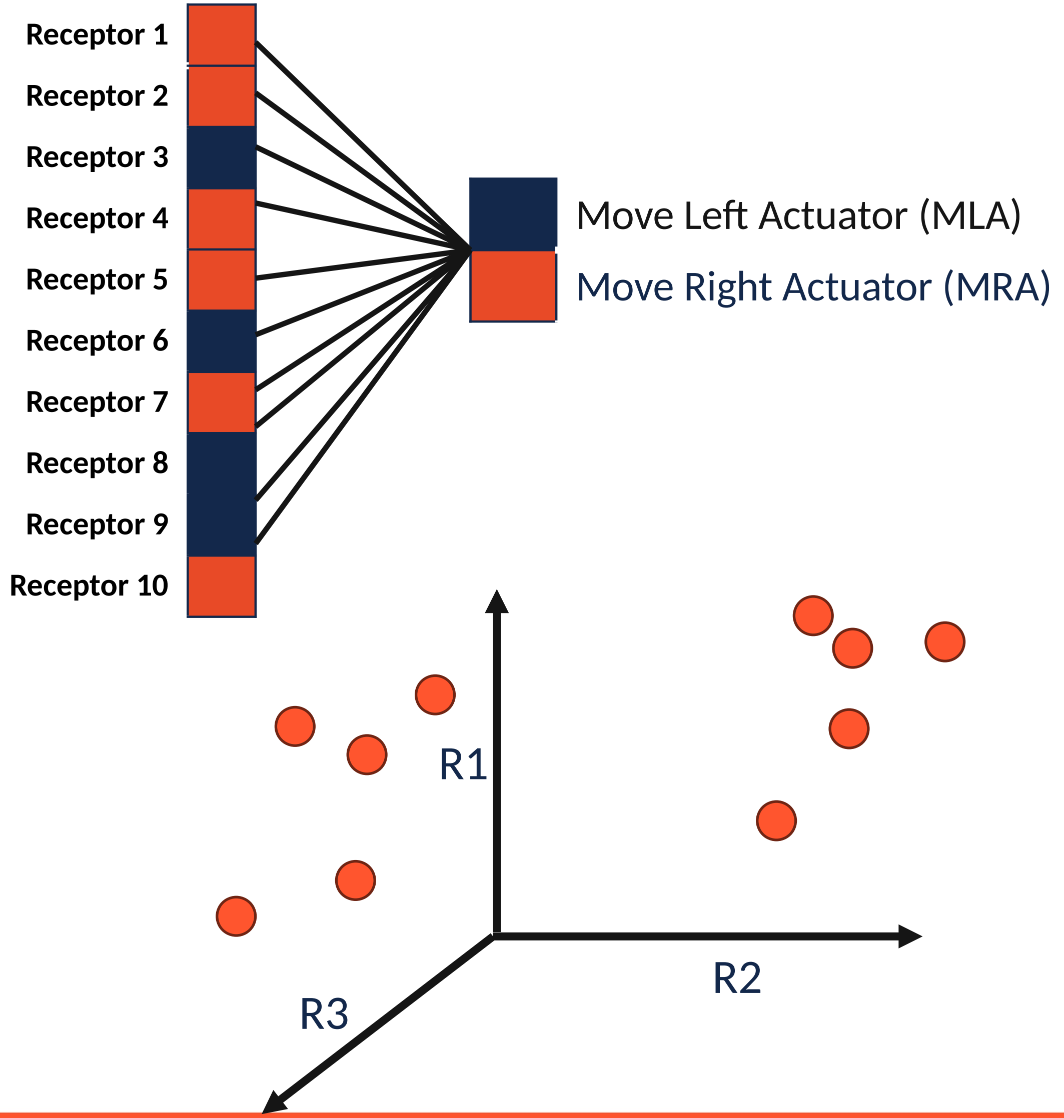
The Pattern Recognition Problem



A Simple Sensory Array

- Each receptor detects one feature
- Input pattern = activity across sensors
- Outputs trigger simple actions

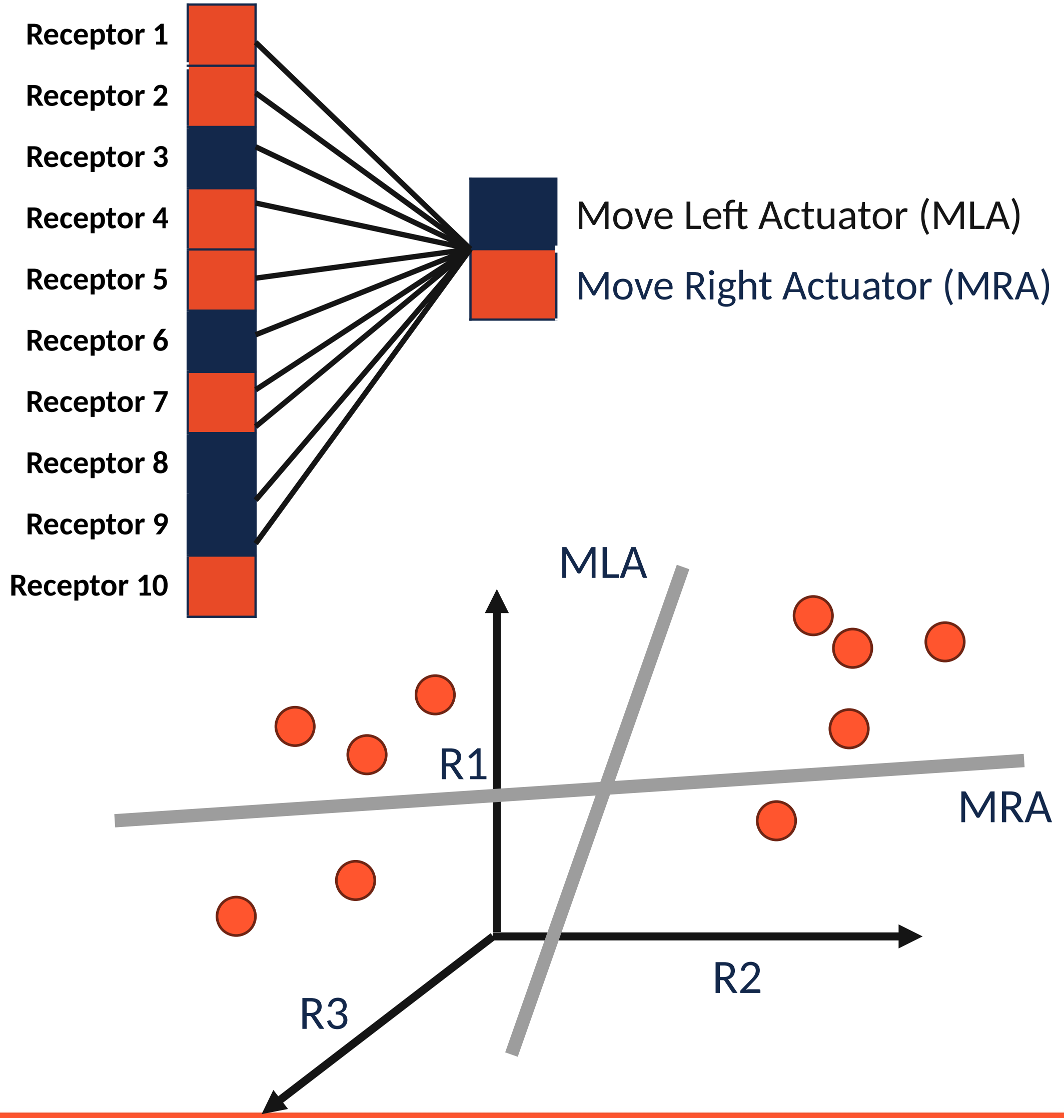
The Pattern Recognition Problem



From Inputs to Feature Space

- Each receptor = one dimension
- Pattern = point in high-dimensional space
- Categories = regions separated by boundaries

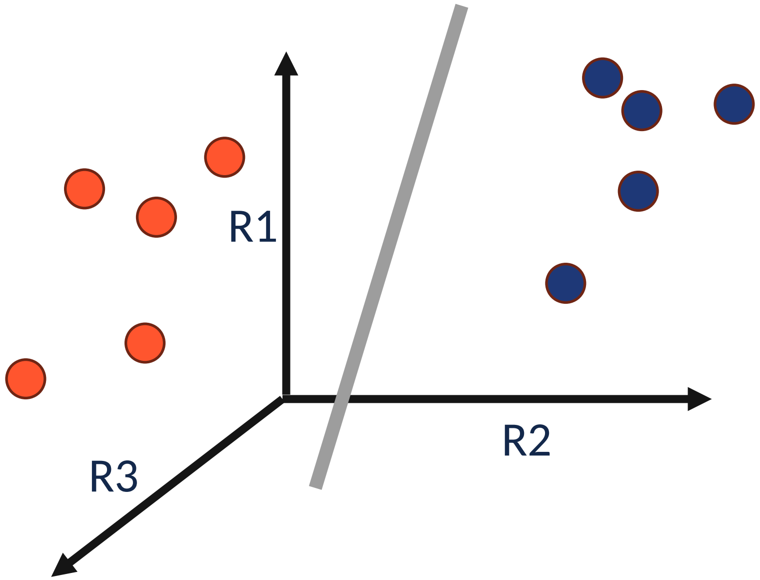
The Pattern Recognition Problem



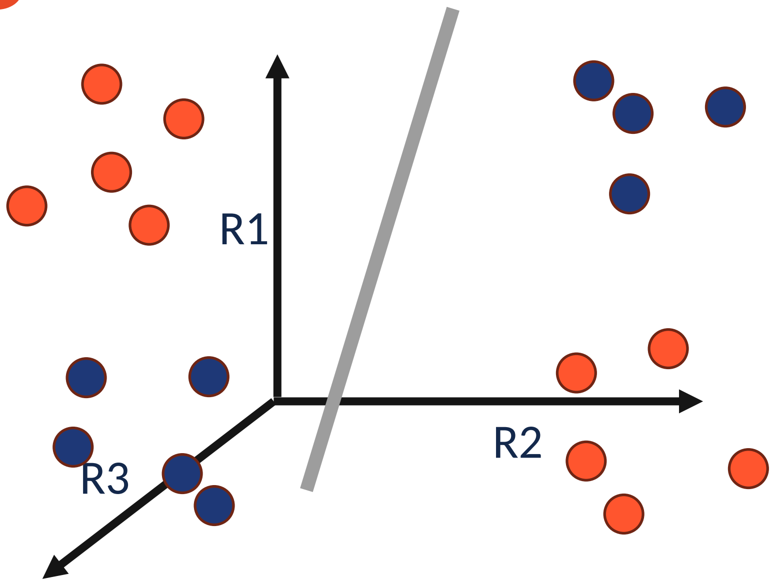
Learning Decision Boundaries

- Weights define importance of each feature
- Threshold divides outputs
- Linear boundary = simple classifier

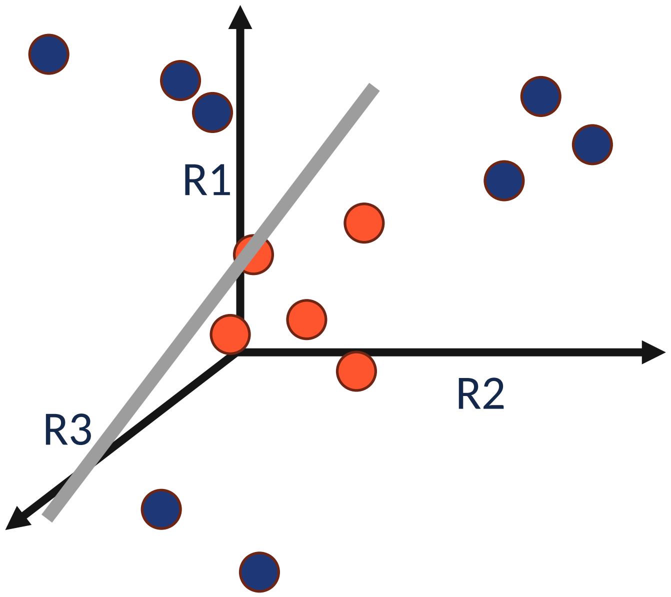
The Pattern Recognition Problem



Linearly Separable



XOR



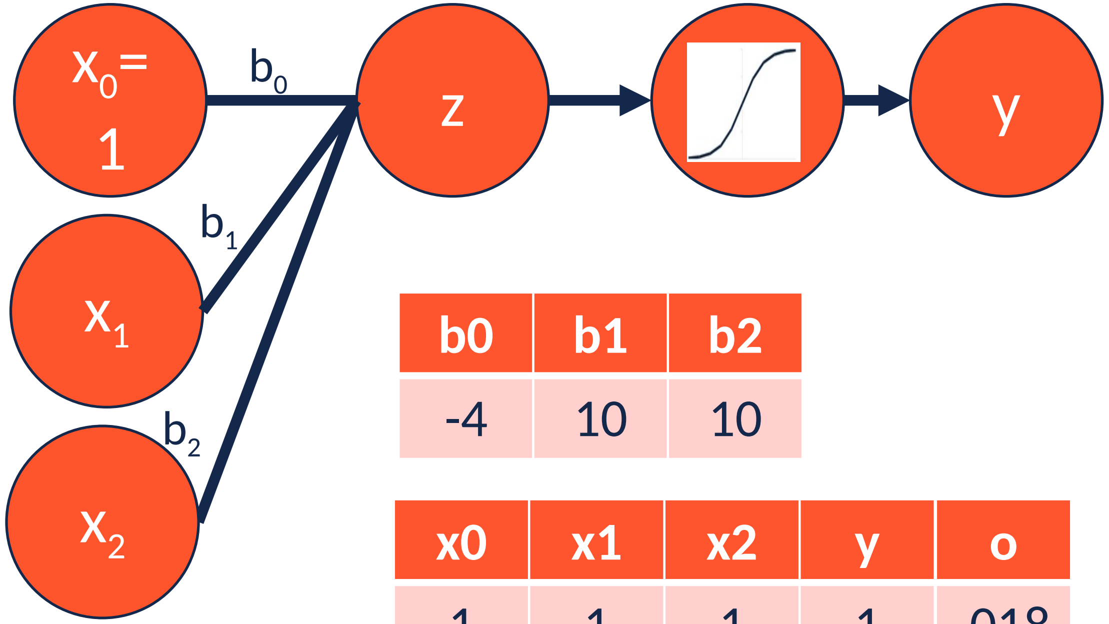
Prototype

When Simple Boundaries Fail

- Overlapping cues blur categories
- Some rules are nonlinear: XOR (A or B but not both) and Prototypes
- Need for multi-layer or hierarchical processing

Logical Computation in Neural Circuits

Logical OR Neural Circuit



b0	b1	b2
-4	10	10

x0	x1	x2	y	o
1	1	1	1	.018
1	1	0	1	.998
1	0	1	1	.998
1	0	0	0	.999

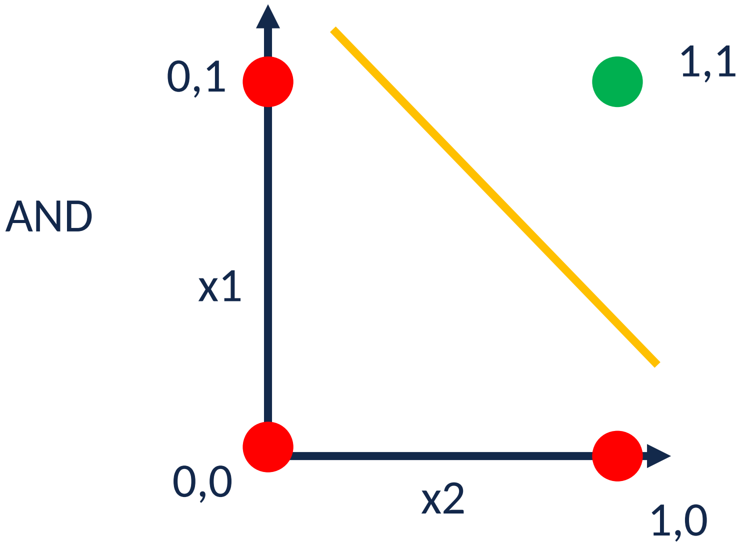
Neuron	b	x	x*b
x_0	-4	1	**-4**
x_1	10	0	**0**
x_2	10	0	**0**
z			**-4**
y=sigmoid(z)			**0.018**

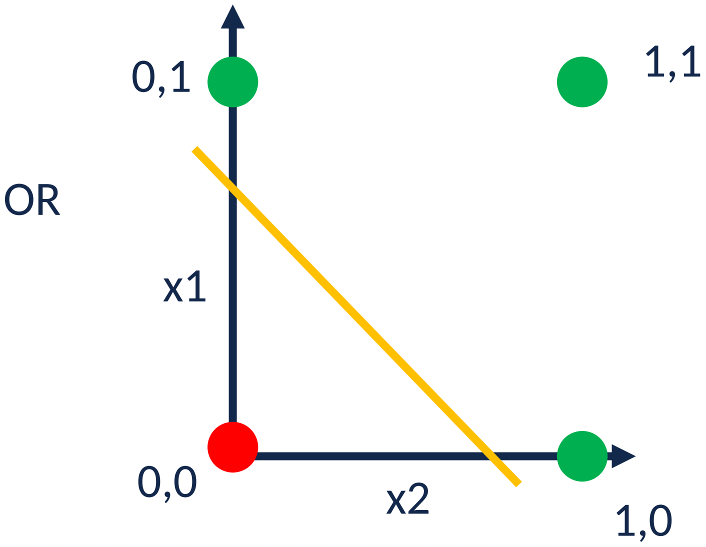
Logical Computation in Neural Circuits

Function	b0	b1	b2
AND	-15	10	10
OR	-4	10	10

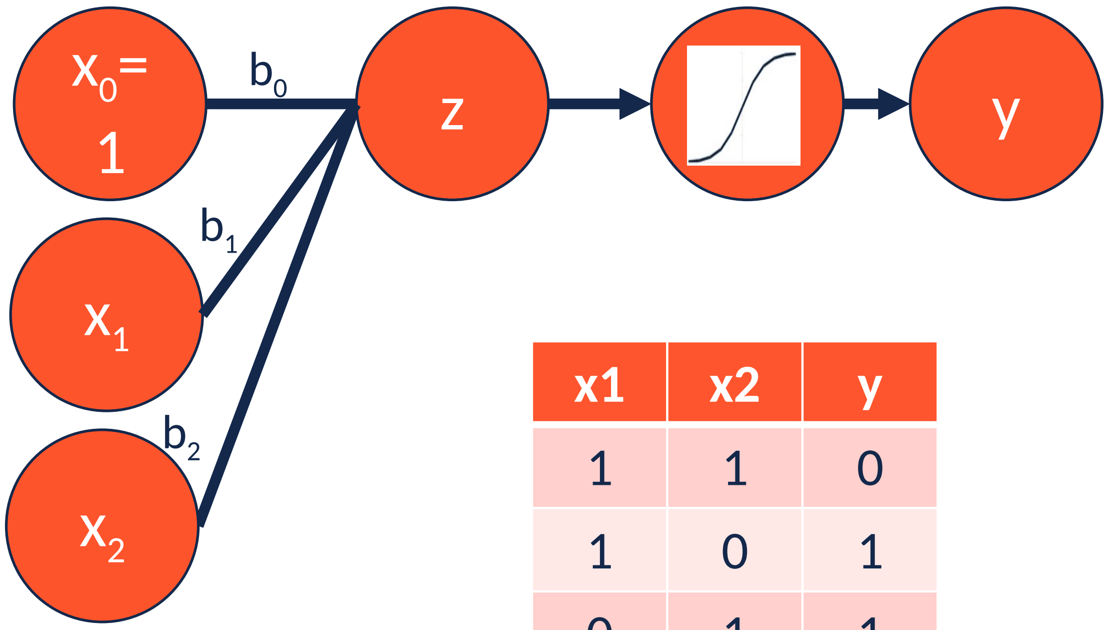
Takeaway: AND vs. OR

- For both functions, both inputs need to have the same positive weights
- But the two functions need to have different bias values
- Bias = neuron's threshold or baseline activity





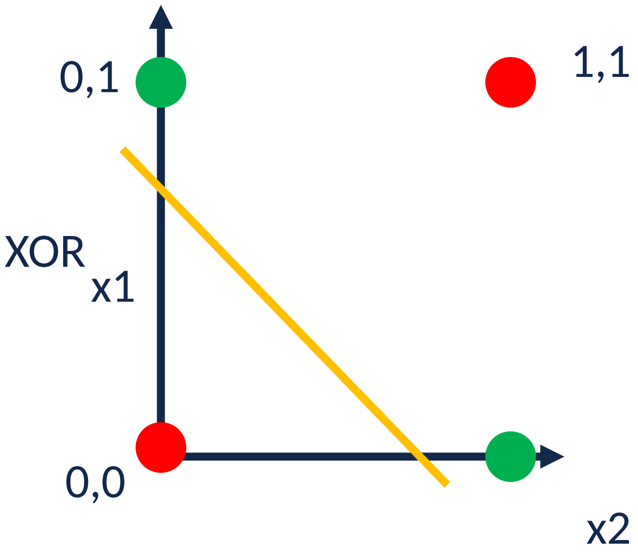
Logical Computation in Neural Circuits

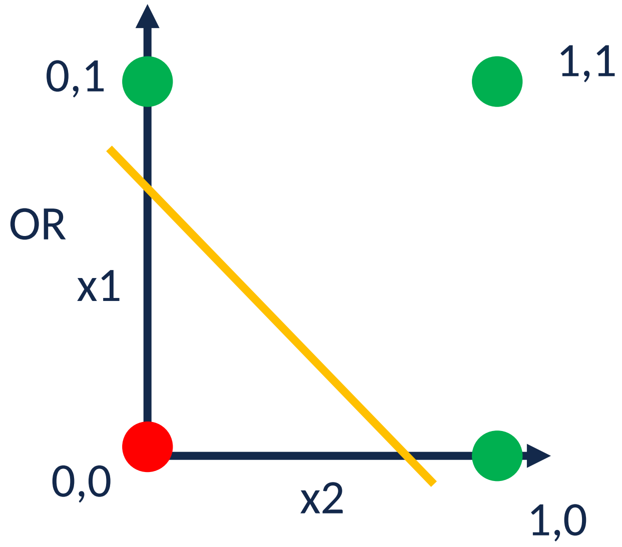


x1	x2	y
1	1	0
1	0	1
0	1	1
0	0	0

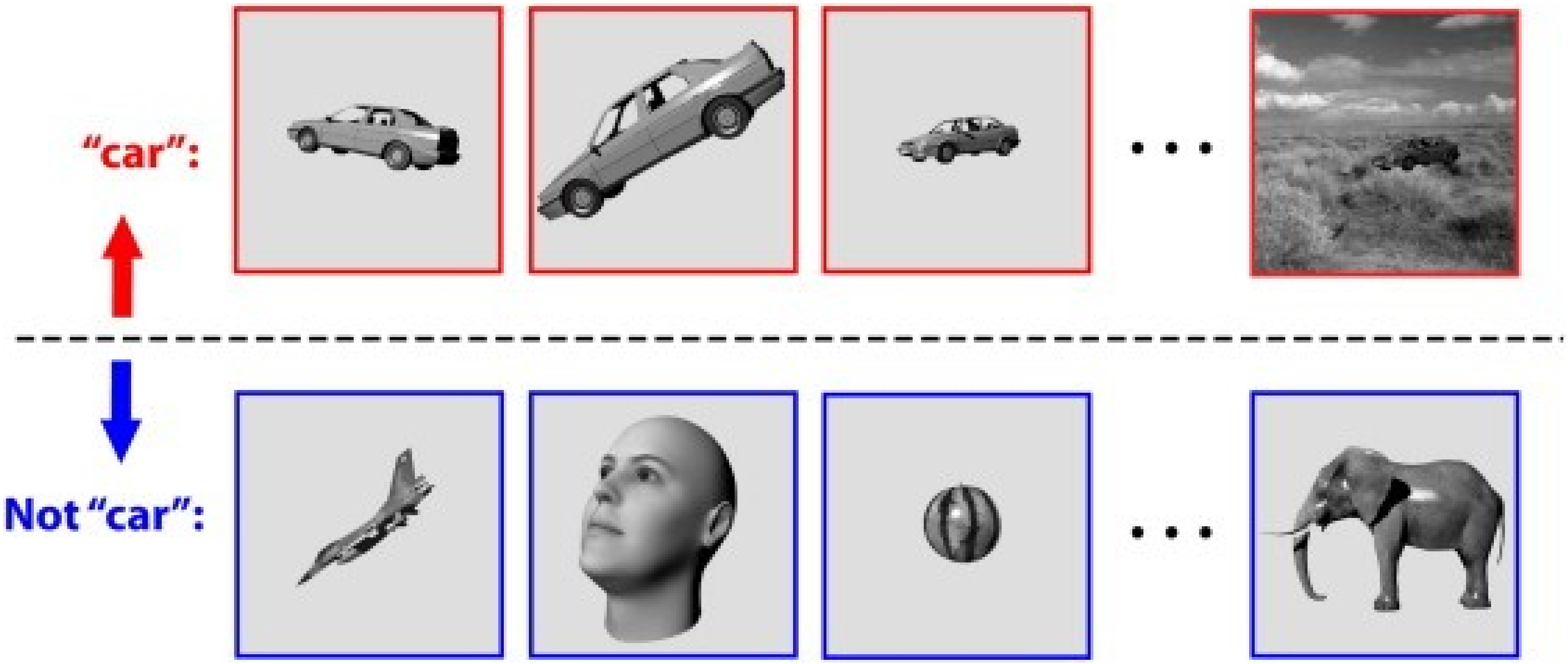
Logical XOR

- XOR = output 1 if inputs are different
- Not linearly separable (can't do with 1 neuron)
- Solution: combine AND, OR, NOT





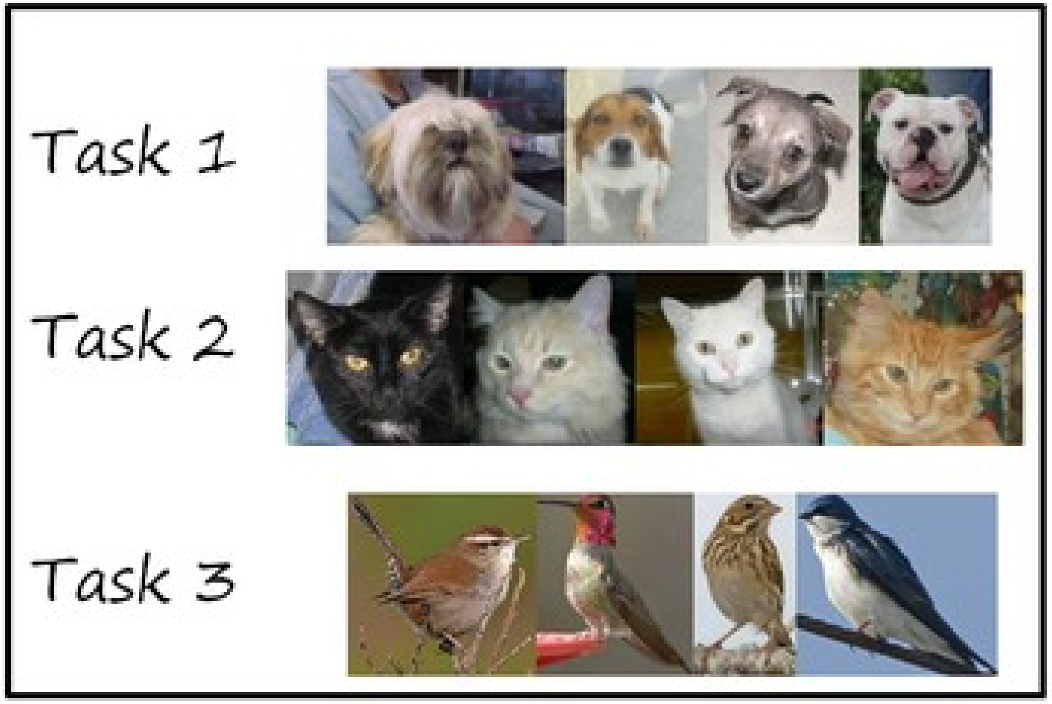
The Pattern Recognition Problem



Invariance Problem

- Same object, different inputs
- Sensory patterns shift with context
- Recognition must extract stable features

The Pattern Recognition Problem



Continual Learning and Interference

- World and experience constantly change
- Learning risks overwriting old knowledge
- Brains balance stability-plasticity dilemma

The Pattern Recognition Problem

Computation
What does the system do?

Algorithm
How does the system do it?

Implementation
How is the system realized?

Summary

- Pattern recognition co-evolved with learning
- Challenges: noise, overlap, nonlinearity, and change

What’s Next:

How biological and artificial systems solve these problems